



# Ahmed Aziz Ben Haj Hmida

Machine Learning Engineer (EPFL, MSc Data Science, GPA 5.3/6) specialising in computer vision, LLMs, and agentic AI systems. Track record of building production-ready ML pipelines with significant, measurable performance gains. Hands-on approach across the full stack from data to deployment.

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## Professional Experience

**Rollomatic SA** Le Landeron, Switzerland  
*Master's Thesis – Curve-Based CNC Tool Similarity Retrieval - **Grade 6/6*** Aug 2025 – Mar 2026

- Designed and deployed an end-to-end CNC tool similarity retrieval system combining geometric curve representations and deep visual embeddings, achieving **11%** improvement in retrieval accuracy within industrial R&D environment.
- Architected scalable indexing and benchmarking pipelines tailored to manufacturing constraints, enabling seamless integration into production R&D workflows.

**Supervisors:** Prof. Jean-Philippe Thiran - jean-philippe.thiran@epfl.ch, Emna Maghrebi - e.maghrebi@rollomatic.ch

**Rollomatic SA** Le Landeron, Switzerland  
*Machine Learning Internship* Feb 2025 – Aug 2025

- Preprocessed and curated a proprietary dataset of 5000+ industrial CNC tools, enabling robust similarity learning.
- Built hybrid retrieval pipelines fusing structured tool parameters with visual embeddings, reducing manual search time by **96%** across the large industrial CNC dataset.

**Supervisor:** Emna Maghrebi - e.maghrebi@rollomatic.ch

## Technical Skills

**Core Expertise:** Python, PyTorch, scikit-learn, NumPy, Pandas

**LLMs & NLP:** HuggingFace Transformers, PEFT, LoRA, RLHF, Quantization

**Agentic AI Systems:** LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, MCP, RAG systems

**Computer Vision:** OpenCV, torchvision, Vision Transformers, CNNs, ResNet, scikit-image

**Data Engineering:** PySpark, PostgreSQL, SQL, FAISS, Milvus, ChromaDB

**MLOps & DevOps:** Docker, Kubernetes, MLflow, W&B, DVC, FastAPI, CI/CD, AWS, GCP

**Visualization:** Matplotlib, Seaborn, Plotly, Power BI

**Other Languages:** Java, SQL, Scala

## Education

**EPFL – École Polytechnique Fédérale de Lausanne** Lausanne, Switzerland  
*M.Sc. in Data Science* Sep 2023 – Mar 2026  
GPA: 5.3/6

*Relevant Coursework: Machine Learning, Natural Language Processing, Computer Vision, Reinforcement Learning, Data Analysis, Database Systems, Data Science for Business*

**Technical University of Denmark (DTU)** Lyngby, Denmark  
*3rd Year of Bachelor - Exchange* Aug 2022 – Jul 2023  
GPA: 5.35/6

**EPFL – École Polytechnique Fédérale de Lausanne** Lausanne, Switzerland  
*B.Sc. in Communication Systems* Sep 2020 – Jul 2023  
GPA: 5.1/6

**Pioneer School of Nabeul** Nabeul, Tunisia  
*Baccalaureate in Mathematics* Sep 2018 – Jun 2019

## Research Projects

**Unsupervised Reverse Knowledge Distillation and Model Merging** Aug 2024 – Feb 2025  
*EPFL LTS4 Lab*

- Designed a reverse distillation framework transferring knowledge from fine-tuned specialist models into larger architectures, preserving general capabilities while boosting niche-task performance.
- Applied model merging techniques (weight interpolation, task vectors) to combine specialised and general models without full retraining.

**Supervisor:** Prof. Pascal Frossard - pascal.frossard@epfl.ch, Ortal Senouf

**Discovering promising new startups through evaluating teams' potentials** Sep 2023 – Dec 2023  
*EPFL ENTIC Lab*

- Applied NLP and multimodal ML to infer MBTI personality types and emotional signals from pitch recordings,

creating behavioural features from unstructured audio/text data.

- Built ensemble classification models predicting startup success likelihood from team dynamics signals, achieving actionable rankings across a labelled cohort of entrepreneurial teams.

Supervisor: Prof. Davide Bavato

## Personal & Academic Projects

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### AI-Powered Food & Nutrition Assistant 🤖

May 2026 - Present

- Built a React frontend and Python backend with a LangChain ReAct agent for natural language meal logging, GPT-4o vision for photo-based meal analysis, and ChromaDB semantic search over 200k+ recipes enriched with USDA nutritional data.
- Tracked daily macro intake with personalised targets (Mifflin-St Jeor), automated email digests via an MCP tool, and deployed to AWS with a GitHub Actions CI/CD pipeline and CloudWatch monitoring.

### LLM Agent-Driven Car Search Assistant 🤖

Mar 2026 - May 2026

- Architected a full-stack conversational search tool using LangGraph + LangChain, enabling users to describe a vehicle in natural language and automatically populate structured queries against a PostgreSQL database.
- Generated ranked listings with match scores, interactive Plotly/Matplotlib charts, and auto-exported PDF comparison reports for saved car profiles.

### LLM Teaching Assistant for MCQ Answering 🤖

Feb 2024 - Jun 2024

- Fine-tuned Qwen1.5 using SFT and DPO with 4-bit quantization and RAG retrieval, balancing response quality and inference efficiency for an educational setting.
- Built an evaluation pipeline using LLM-as-a-Judge and cross-validation to measure pedagogical accuracy, enabling systematic model comparison and iteration.

### Coin Detector and Counter 🤖

Feb 2024 - Jun 2024

- Built a multi-stage vision pipeline combining CNN-based segmentation and ResNet50 / ViT classification to detect and identify coins across varied and uncontrolled backgrounds.

### RL for Motor Skill Transfer 🤖

Feb 2024 - Jun 2024

*EPFL Alexander Mathis Lab*

- Trained an expert RL agent on the Myosuite baoding balls task and distilled its motor control strategies to novice agents via behaviour cloning, improving sample efficiency and final performance on a complex continuous-control benchmark.

### Machine Learning Operations 🤖

Jan 2023 - Mar 2023

- Built a production-grade ML pipeline integrating PyTorch, Docker, Kubernetes, MLflow, DVC, AWS, and GCP — covering model versioning, containerised training, and automated deployment.

## Certificates

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**IBM RAG and Agentic AI** – IBM

Apr 2026 - Jun 2026

**DevOps and AI on AWS** – AWS

May 2026 - Jun 2026

**Structuring Machine Learning Projects** – DeepLearning.AI

Sep 2023 - Oct 2023

**Certificate in Artificial Intelligence** – GOMYCODE

Jun 2023 – Sep 2023

## Leadership and Volunteering

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**EPFL Coaching** – CMS section Coach

Sep 2023 - Jun 2024

**Rotary International** – Member

Jul 2018 - Apr 2019

**Tunisian Red Crescent** – Volunteer

May 2018 - Apr 2019

## Languages

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Arabic (Native), French (Native/C2), English (C1), German (A2)